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Management of Aquatic Resources in Large Catchments: Recognizing Interactions Between Ecosystem Connectivity and Environmental Disturbance

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Abstract

Management within catchment basins must be approached with an empirically based understanding of the natural connectivity and variability of structural and functional properties of riverine ecosystems. Rivers are four-dimensional environments involving processes that connect upstream-downstream, channel-hyporheic (groundwater), and channel-floodplain (riparian) zones or patches, and these differ temporally. Natural and human disturbances, including biotic feedback (such as predation, parasitism, and other food web dynamics), interact to determine the most probable biophysical state of the catchment ecosystem. Human disturbances can be quantitatively determined by deviations from an observed biophysical state (baseline), but usually this requires long-term ecological data sets. A case history of the Flathead River-Lake system in Montana (USA) and British Columbia (Canada) is summarized to illustrate how disturbances interact at the catchment level of organization. Owing to the natural complexities of catchment ecosystems and the cumulative effects of human disturbances, the rationale and logistics of obtaining long-term data often seem intractable and excessively expensive. The naive alternative is to derive and implement simplistic procedures that are agency specific and often result in management actions that interfere with each other. We argue that integrated management at the catchment level is needed and propose some simple principles, beginning with broader based collegiate training for prospective managers.

Key words. Ecosystem, river, catchment, drainage basin, management, disturbance, natural resources, watershed, Flathead River, Montana.

Introduction

Professor Noel Hynes first synthesized the concept of ecological connectivity in the context of river systems in his Baldi Lecture at the 19th Congress

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of the International Society for Pure and Applied Limnology (Hynes 1975). He eloquently described how rivers are a manifestation of the biogeochemical nature of the valleys they drain, and he proposed that understanding the inherent connectivity between terrestrial and lotic biotopes would lead to important predictions about the future structure and function of river ecosystems.

In the nearly two decades since that seminal lecture, several paradigms (reviewed by Cummins et al. 1984) emerged from scores of studies that examined spatial and temporal aspects of geomorphic, hydrologic, thermal, and riparian influences on biotic attributes (e.g., diversity, zonation, food web associations, bioproduction) of rivers. The river continuum concept (Vannote et al. 1980, Minshall et al. 1985) provided a template for examining how biotic attributes of rivers change within the longitudinal gradient from headwaters to ocean confluence. The serial discontinuity concept (Ward and Stanford 1983*a*) provided a construct for the propensity of rivers to predictably reset biophysical attributes in relation to distance downstream from on-channel impoundments. Comparison of organic matter budgets in streams in different biomes provided the basis for the riparian control concept and demonstrated the extreme importance of allochthonous debris (wood and leaves) in lotic systems (Cummins et al. 1984, Harmon et al. 1986, Webster and Benfield 1986, Ward et al. 1990, Gregory et al. 1991). The nutrient spiraling concept (Webster and Patten 1979, Newbold et al. 1983) led to an understanding of how plant growth nutrients are transformed from dissolved to particulate states during translocation from upstream to downstream reaches. Lastly, the ecotone concept (Naiman and Décamps 1990, Holland et al. 1991) has fostered greater understanding of the extreme importance and potential predictive power related to transformations and fluxes of materials that occur within boundaries between functionally interconnected patches that form the riverine landscape. In many ways the ecotone concept integrates the other paradigms by emphasizing the functional connectivity inherent in all ecosystems.

Studies articulating these paradigms and other syntheses of stream ecology (Lock and Williams 1981, Barnes and Minshall 1983, Dodge 1989, Stanford and Covich 1988, Yount and Niemi 1990), plus a great number of other research projects, have largely verified Hynes's proposition that the streams are tightly coupled with catchment characteristics. Drainage basins or catchments (i.e., the river valley in Hynes's context) may indeed be characterized as ecosystems composed of a mosaic of terrestrial "patches" (Pickett and White 1985) that are connected (drained) by a network of streams. Of course, the lotic environment itself is a smaller scale patchwork or mosaic of habitats in which materials and energy are transferred (connected) through dynamic, biodiverse food webs. In most catchments, on-channel lakes and floodplain aquifers dramatically increase the complexity of the ecosystem, in contrast